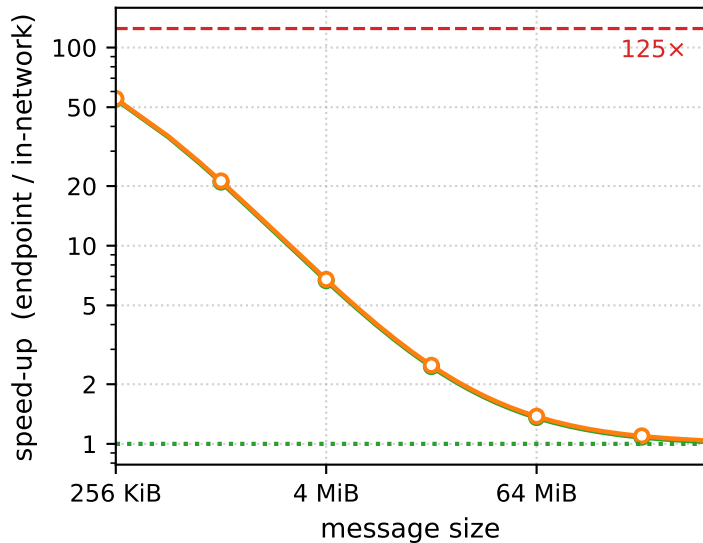
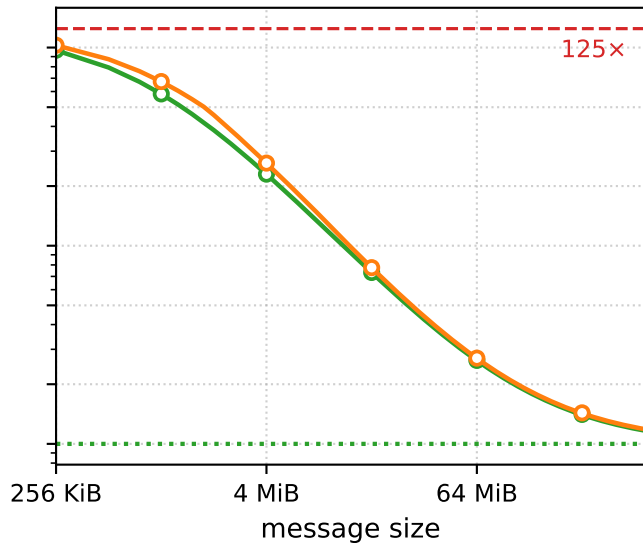


(a) single switch,  $d = 1$



(b) three-tier,  $d = 3$



— Reduce-Scatter, model  
○ Reduce-Scatter, measured

— AllGather, model  
○ AllGather, measured

--- latency bound  
..... bandwidth reference